

Logisim: A User-Friendly Logic Simulator

Logisim not only lets you simulate your logical circuits but also edit circuits while simulation is running. Therefore you can immediately rectify flaws without the need of separate edits and loading

Using logic simulator software tools, electronics designers can digitally interpret the behaviour of circuits they create. This offers a great way to ensure accuracy of their design and correctness of the applied logic before jumping into the actual production. One such popular logic simulation software is Logisim.

Logisim is a free and open source software compatible with multiple platforms like Windows, MacOS and Linux. The ease of Logisim comes from its interactive graphical user interface developed with Java Swing GUI library. Users can create circuits with simple drag actions of the mouse. Logisim enables them to create circuits of any scale, including sub-circuits as well as complex structures. Colour-coded wires impart clarity to the representation.

Using the software

Logisim is simple, compact and portable. The installation process is easy and startup takes only a few seconds. Logisim opens as a simple workspace with panels on the top and the left. The workspace is dotted. You can drag logic gates, components and wires and connect them inside the workspace to create your circuit.

Connecting the components together is simple. Simply dragging the mouse between the intended

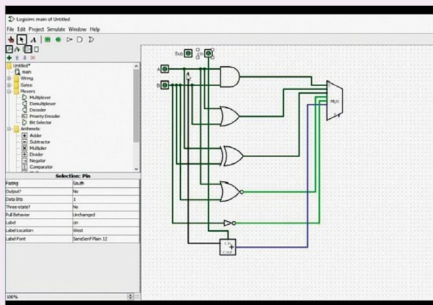


Fig. 1: The Logisim interface

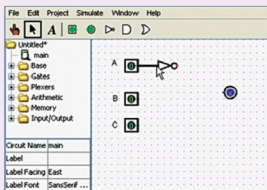


Fig. 2: Connecting wires in Logisim

components joins them with a digital wire. The wires can run horizontally and vertically.

All the elements of the circuit are colour-coded. For example, input ports are represented in green and output ports in blue. Also, active wires during simulation turn bright green, while wires with no electricity remain deeper green by default. Any flawed wiring is indicated in

red. All input and output ports have a designated value of either 0 or 1 assigned to them, indicating their active status.

The upper edge of the software contains two menu bars. The topmost bar contains File, Edit, Project, Simulate, Windows and Help menus. You can perform different actions from these menus, such as open or continue a project,

edit a circuit, access tutorials or run a simulation. Project menu contains many sub-menus like Add Circuit, Load or Unload Library, View Simulation Tree, Analyse Circuit and so on. Under Analyse Circuit, you can check values of the input, output and truth table that the logic circuit follows.

The other toolbar consists of some useful tools and shortcuts. Finger icon lets you toggle between a cursor